Doppler Effect

DISCLAIMER

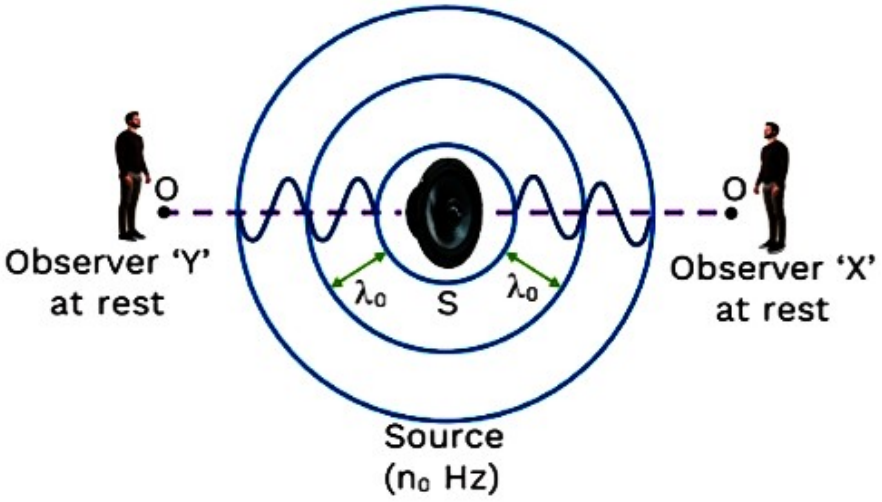
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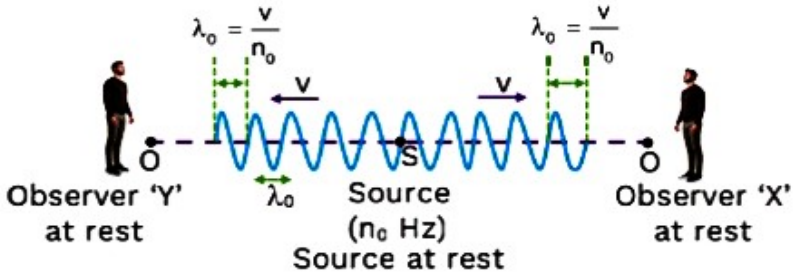
Doppler Effect

Doppler's Effect

Variation in frequency of sound heard by an observer from frequency produced by source depends on relative motion of source of sound and observer along the line joining them. If the observer are moving away from each other observed frequency will be less than actual frequency and in case if they are moving towards each other then observed frequency will be greater than actual frequency.

(a) When both source and observer are stationary: In this case there would be no change in wavelength or frequency by sound produced source in air.

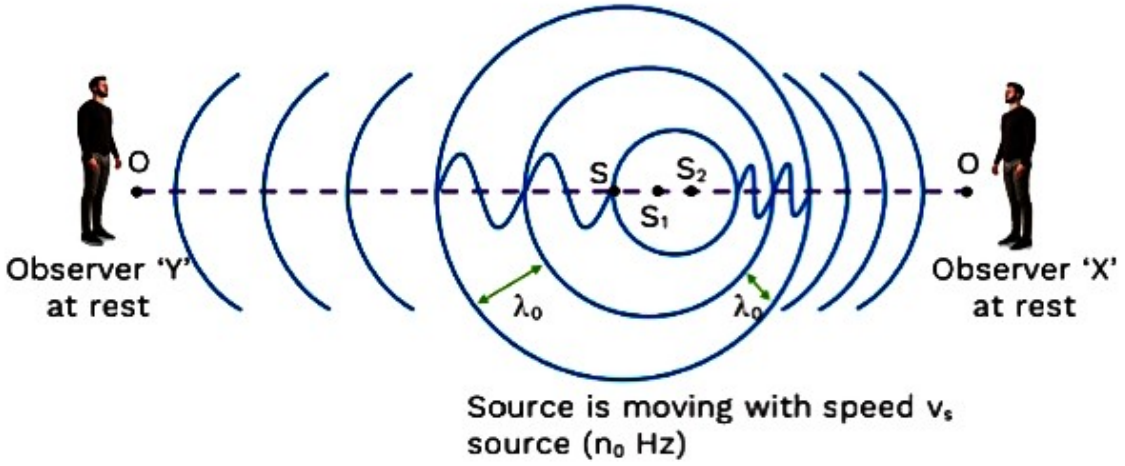


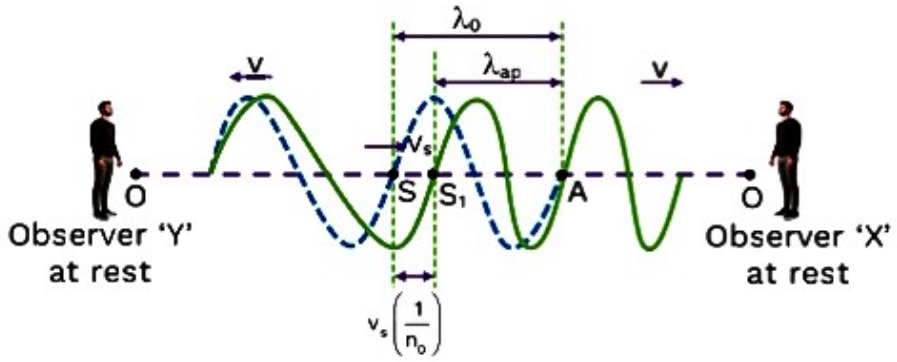


$$\lambda = \frac{v}{f_0} = \lambda_0$$

$$f = \frac{v}{\lambda_0} = f_0$$

(b) Sound Source is Moving, and Observer is Stationary: If source having frequency f traveling with velocity v_s along the line joining source and observer then their will be a change in wavelength produced by source





Here wavelength that is created by source in medium get changed.

$$\lambda' = vT - v_sT$$

$$\lambda' = \frac{v - v_s}{f} \qquad \text{(for observer X)}$$

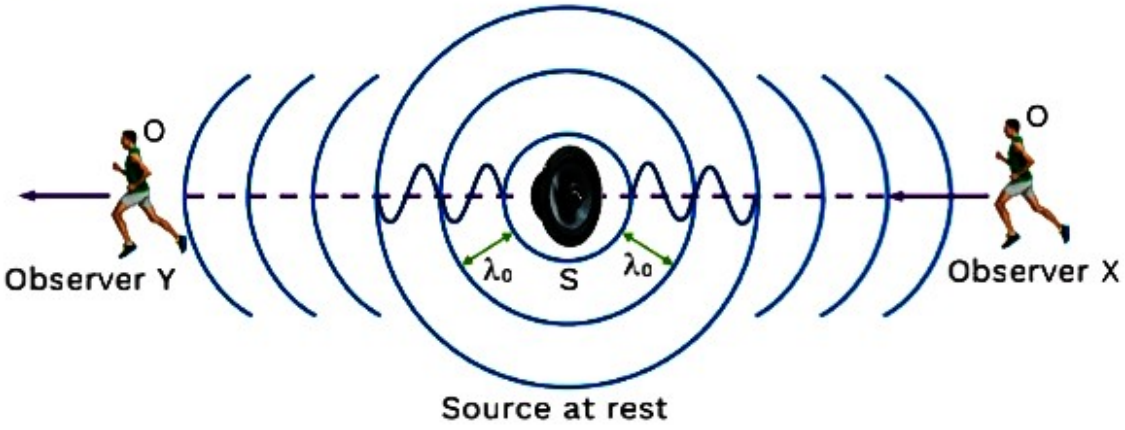
or $$\lambda'' = \frac{v + v_s}{f} \qquad \text{(for observer Y)}$$

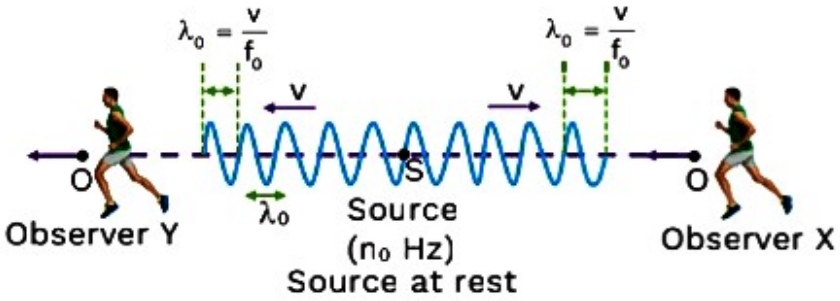
So apparent frequency, $f' = \frac{v}{\lambda'}$

$$\Rightarrow \boxed{f' = f\left(\frac{v}{v \pm v_s}\right)} \qquad \begin{bmatrix} + \text{ for observer X} \\ - \text{ for observer Y} \end{bmatrix}$$

(c) Sound Source is Stationery and Observer is Moving with Velocity v_0 Along the Line Joining Them:

Wavelength of sound produced by a stationary source will remain same but there will be a change in frequency as velocity of sound with respect to observer changes because of its motion





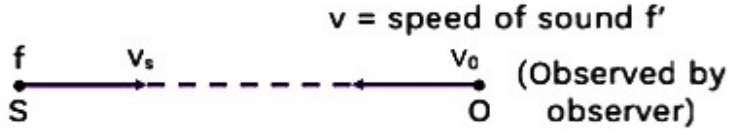
time required to cross one wavelength,

$$T' = \frac{\lambda_0}{v + v_0} \qquad \text{(for observer X)}$$

$$\Rightarrow \boxed{f' = \frac{f(v \pm v_0)}{v}} \qquad \begin{bmatrix} + \text{ for observer X} \\ - \text{ for observer Y} \end{bmatrix}$$

(d) The Source and Observer Both are Moving with Velocities v_s and v_0 Along the Line Joining Them:

If both source of sound and detector moves in such a way that separation between them changes with time, then frequency observed by the detector will be different from actual.

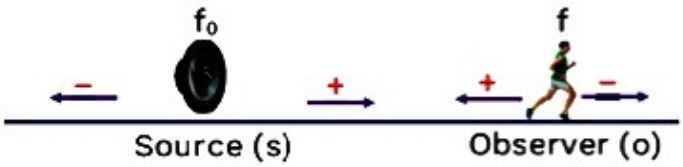


$$f' = f\left(\frac{v + v_0}{v - v_s}\right)$$

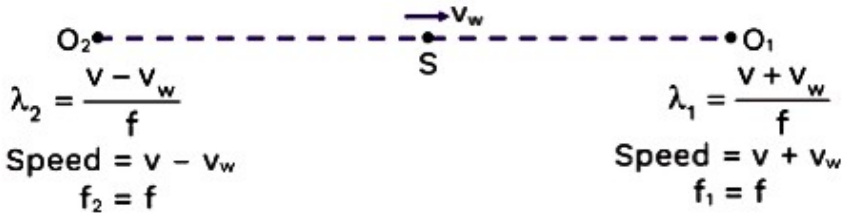
This formula can be used for all the cases by following the sign convention given below.

Sign convention:

In this formula sign of observer velocity and source velocity is taken positive only in case they are moving toward each other otherwise velocity of source and observer will be negative.



Medium is Moving



If medium is moving, then speed of sound w.r.t ground will be difference.

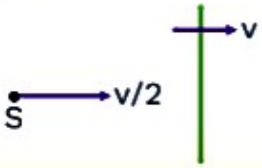
For the sound going with the wind, speed w.r.t t ground will be $v + v$, and if the sound is going against the wind, then speed or sound w.r.t ground will be $v - v_{wind}$.

Expression for frequency is given by

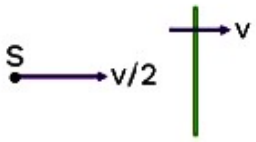
$$f' = f\left(\frac{v \pm v_w + v_0}{v \pm v_w - v_s}\right)$$

sign of velocity of observer and source is as given in previous section.

Q **A wall and a source of sound is moving in same direction with velocity v and v/2 respectively. Find the ratio of incident sound wavelength to the reflected sound wavelength by the wall assuming velocity of sound to be 10 v:**



Sol. $f'_{wall\,(received)} = \frac{10v - v}{10v - v/2} f = \frac{9v}{9.5v} f \Rightarrow \lambda_1 = \frac{9.5v}{f}$



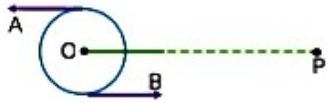
$$f'' = \frac{10\,v}{10\,v + v} f' = \frac{10\,v}{11\,v} \times \frac{9\,v}{9.5\,v} f \Rightarrow \lambda_2 = \frac{11v \times 9.5}{9f}$$

$$\Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{9.5}{11 \times 9.5} \times 9 = \frac{9}{11}$$

Q **A whistle having frequency 480 Hz moving in a circle of radius 2 m with angular velocity of 15 rad/s. Find lowest and highest frequency heard by a listener standing at rest far away from the source (velocity of sound is 330 m/s)**

Sol. Maximum linear speed of sound source with respect to observer of sound source ωR (v_s)

Maximum frequency is given by



$$f_{max} = f\left(\frac{v + v_0}{v - v_s}\right) = \left(\frac{v}{v - \omega R}\right) f = \frac{330}{330 - 30} \times 480 = 528 \text{ Hz}$$

Minimum frequency is given by

$$f_{min} = f\left(\frac{v + v_0}{v - v_s}\right) = \left(\frac{v}{v + \omega R}\right) f = \frac{330}{330 + 30} \times 480 = 440 \text{ Hz}$$

Q **A motorist and a band party is moving towards a ball with speed v_m and v_b respectively if motorist is behind band party and speed of sound is v than find beat frequency heard by the motorist.**

Sol. Actual frequency of band music is f. The direct frequency heard by motorist is



$$f_1 = \left(\frac{v + (v_m)}{v - (-v_b)}\right) f = \left(\frac{v + v_m}{v + v_b}\right) f$$

Frequency of incident sound source on wall will be

$$f' = \left(\frac{v + 0}{v - v_b}\right) f = \left(\frac{v}{v - v_b}\right) f$$

This frequency will also be reflected by the wall and act is a second sound source two motorist

$$f_2 = \left(\frac{v + (v_m)}{v - 0}\right) f' = \left(\frac{v + v_m}{v}\right) \times \left(\frac{v}{v - v_b}\right) f = \left(\frac{v + v_m}{v - v_b}\right) f$$

Therefore, the beat frequency is

$$n = f_2 - f_1 = \left(\frac{v + v_m}{v - v_b}\right) f - \left(\frac{v + v_m}{v + v_b}\right) f$$

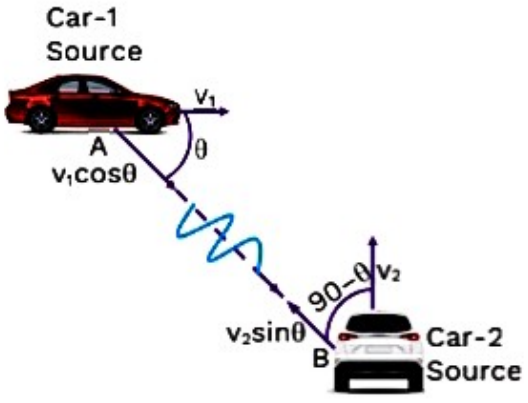
$$= \frac{(v + v_b) - (v - v_b)}{v^2 - v_b^2} (v + v_m) f$$

$$\Rightarrow \quad n = \frac{2v_b (v + v_m) f}{v^2 - v_b^2}$$

Doppler's Effect When Source and Observer are not in Same Line of Motion

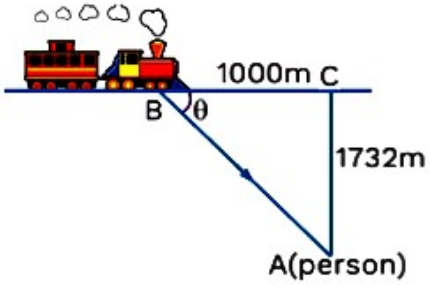
Let us consider the situation shown in figure. Two cars 1 and 2 are moving along two roads crossing perpendicularly at speeds v_1 and v_2. When car-1 sounds a horn of frequency n_0, it emits sound pulse and car-2 receives the sound at the position as shown in the figure. The direction of sound is from source position to observation position. It means Doppler Effect is applicable along the line AB. In such cases, we use velocity components of the cars along the line joining the source and observer thus the apparent frequency of sound heard by car-2 can be given as

$$f_{ap} = f_0 \left[\frac{v + v_2 \cos(90 - \theta)}{v - v_1 \cos\theta}\right] = f_0 \left[\frac{v + v_2 \sin\theta}{v - v_1 \cos\theta}\right]$$



Q **A train moving with the speed 72 km/hr producing a sound having frequency 640 Hz is heard by a portion sitting at 1732 m from point c as shown in figure. Find frequency heard by the person if speed of sound is 330 m/s.**

Sol. Speed of moving train $= \frac{5}{18} \times 70 = 20$ m/s



From the given figure $\tan\theta = \frac{1732}{1000} = \sqrt{3}$,

$\Rightarrow \quad \theta = 60°$

Taking the component of velocity of source along the given direction observed frequency can be

$$f = \frac{v}{v - u\cos\theta}, \; f = \frac{330}{330 - 10} \times 640\,\text{Hz} = 660 \text{ Hz}.$$